CHAPTER 1

ELECTRIC CHARGES AND FIELDS

CHAPTER ESSENCE AT A GLANCE:
The fundamental physics of static electric charges, Coulomb's force law, dipole fields, and Gauss's law flux theorems governing charge distributions.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Coulomb's Law & Superposition:** Electrostatic force acts along the line joining two stationary point charges, directly proportional to the product of charges and inversely proportional to the square of separation distance. The net force on any charge is the vector sum of individual forces.

• **Electric Dipole & Dipole Moment:** Equal and opposite charges separated by small distance 2a form a dipole with moment vector pointing from negative to positive. The axial electric field is twice the equatorial field at large distances.

• **Dipole in Uniform Electric Field:** Experiences zero net translational force in a uniform field, but experiences a deflecting torque tending to align it with the external field lines.

• **Gauss's Law & Charge Symmetry:** The total outward electric flux through any closed Gaussian surface equals the net enclosed charge divided by the permittivity of free space, proving highly effective for symmetric charge distributions.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}$$

Coulomb's Law in free space with dielectric medium scaling

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r_{12}^2}\hat{\mathbf{r}}_{21} = -\mathbf{F}_{21}$$

Vector form obeying Newton's Third Law

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0}\frac{2p}{r^3} \quad \text{and} \quad E_{\text{eq}} = \frac{1}{4\pi\epsilon_0}\frac{p}{r^3}$$

Dipole electric fields (r >> a): axial field is twice equatorial field

$$\tau = \mathbf{p} \times \mathbf{E} \rightarrow \tau = pE\sin\theta$$

Torque on dipole in uniform electric field with zero net force

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Electrostatic field inside a uniformly charged conducting spherical shell is IDENTICALLY ZERO at all internal points.

• **Alert:** Electric field lines never form closed loops because electrostatic fields are conservative.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Coulomb's Law in vector form and principle of superposition [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Principle: Coulomb's Law & Superposition]** The electrostatic force between two point charges q_1 and q_2 at rest in vacuum separated by distance r is directed along the line joining them and given by F = (one-fourth (0.25)$\pi\varepsilon_0$) · (q_1 q_2 divided by r^2).

• **[Step 1: Vector Formulation]** Force on q_1 exerted by q_2: F12 = (one-fourth (0.25)$\pi\varepsilon_0$) (q_1 q_2 divided by r_{12}^2) unit vector r_{21}, where unit vector r_{21} = r12 divided by r_{12} is the unit vector pointing from q_2 to q_1.

• By Newton's third law, F21 = - F12. [1.5 Marks]

• **[Step 2: Principle of Superposition]** The net force on any charge q_1 due to an assembly of n-1 charges is the vector sum of individual two-body Coulomb forces: F1 = F12 + F13 + ... + F1n = (q_1 divided by 4$\pi\varepsilon_0$) sum from i=2 to n (q_i divided by r_{1i}^2) unit vector r_{i1}. [1.5 Marks]

• **Key Exam Result:** F12 = (one-fourth (0.25)$\pi\varepsilon_0$) (q_1 q_2 divided by r^2) unit vector r_{21} and F net = sum of Fi

2. Electric field on axial line of dipole [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Diagram Schematic]** Electric dipole consisting of -q at (-a, 0) and +q at (+a, 0), dipole length 2a, dipole moment p = q(2a) unit vector i. Field point P is on the axial line at distance r from center O (r > a).

• **[Step 1: Individual Field Magnitudes]** E_+ (due to +q) = (one-fourth (0.25)$\pi\varepsilon_0$) · q divided by $(r - a)^2$ (directed along +p). E_- (due to -q) = (one-fourth (0.25)$\pi\varepsilon_0$) · q divided by $(r + a)^2$ (directed opposite to p). [1 Mark]

• **[Step 2: Vector Sum & Algebraic Reduction]** E axial = E_+ - E_- = (q divided by 4$\pi\varepsilon_0$) [1 divided by $(r - a)^2$ - 1 divided by $(r + a)^2$] = (q divided by 4$\pi\varepsilon_0$) [$((r+a)^2 - (r-a)^2)$ divided by $(r^2 - a^2)^2$] = (q divided by 4$\pi\varepsilon_0$) [4ra divided by $(r^2 - a^2)^2$] = (one-fourth (0.25)$\pi\varepsilon_0$) · [2(q · 2a) r divided by $(r^2 - a^2)^2$]. [1 Mark]

• **[Step 3: Short Dipole Limit (r >> a)]** For r >> a, $(r^2 - a^2)^2$ approx. r^4. Therefore, E axial = (one-fourth (0.25)$\pi\varepsilon_0$) · (2p divided by r^3). In vector notation: E axial = (one-fourth (0.25)$\pi\varepsilon_0$) (2vec(p) divided by r^3). [1 Mark]

• **Key Exam Result:** E axial = (one-fourth (0.25)$\pi\varepsilon_0$) · (2vec(p) divided by r^3) (along dipole axis)

3. Electric field on equatorial line of dipole [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Diagram Schematic]** Dipole of charges -q at A(-a) and +q at B(+a). Field point P is on equatorial line at distance r from center O. Distance from both charges AP = BP = sqrt($r^2 + a^2$).

• **[Step 1: Component Resolution]** Magnitudes: E_+ = E_- = (one-fourth (0.25)$\pi\varepsilon_0$) · q divided by $(r^2 + a^2)$. Vertical components perpendicular to dipole axis: E_+ sin θ and E_- sin θ are equal and opposite, canceling out. Horizontal components parallel to dipole axis add up opposite to p: E eq = 2 E_+ cos θ. [1.5 Marks]

• **[Step 2: Geometric Substitution & Final Expression]** From triangle OBP, cos θ = a divided by $\sqrt{(r^2 + a^2)}$. E eq = 2 · [(one-fourth (0.25)$\pi\varepsilon_0$) q divided by $(r^2 + a^2)$] · [a divided by $(r^2 + a^2)$^(half (0.5))] = (one-fourth (0.25)$\pi\varepsilon_0$) · (q · 2a) divided by $(r^2 + a^2)$^(3 divided by 2) = (one-fourth (0.25)$\pi\varepsilon_0$) · p divided by $(r^2 + a^2)$^(3 divided by 2). For short dipole (r >> a): E eq = (one-fourth (0.25)$\pi\varepsilon_0$) · (p divided by r^3). In vector form: E eq = - (one-fourth (0.25)$\pi\varepsilon_0$) (p divided by r^3). [1.5 Marks]

• **Key Exam Result:** E eq = - (one-fourth (0.25)$\pi\varepsilon_0$) (p divided by r^3); note E axial = 2 · E eq

4. Torque on electric dipole in uniform field [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Force & Couple Analysis]** Dipole with charges +q and -q placed in uniform electric field E at angle θ. Force on +q is F_+ = +q E; force on -q is F_- = -q E. Net translational force F net = F_+ + F_- = 0. Hence no linear acceleration. [1 Mark]

• **[Step 2: Torque Derivation]** The forces act along different lines of action separated by perpendicular distance d = 2a sin θ, forming a couple. Torque magnitude τ = Force × perpendicular distance = (q E)(2a sin θ) = (q · 2a) E sin θ = p E sin θ. In vector cross-product form: τ = p × E. Direction given by right-hand thumb rule. [1 Mark]

• **Key Exam Result:** τ = p × E; τ max = pE at θ = 90°, τ = 0 at θ = 0°, 180°

5. Electric flux and statement of Gauss's Law [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Electric Flux Definition]** Electric flux Φ E through a surface is the total measure of electric field lines passing normally through that surface. Mathematical definition: $\Phi E = \int S\ E \cdot dA = \int S\ E \cos \theta\ dA$. SI Unit: N m^2 C^{-1} or V m. Scalar quantity. [1 Mark]

• **[Step 2: Gauss's Law Statement]** Gauss's Law states that the total outward electric flux through any closed Gaussian surface enclosing volume V is equal to (1 divided by ε_0) times the net electric charge enclosed inside the surface: closed integral S E · dA = q enclosed divided by ε_0. Valid for any arbitrary closed surface and arbitrary charge distribution. [1 Mark]

• **Key Exam Result:** closed integral E · dA = q in divided by ε_0; SI Unit of Φ: N m^2 C^{-1} or V m

6. Quantization of charge and numerical problems [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula & Principle]** Quantization of charge states q = ± n e, where n is an integer and e = 1.602 × 10^{-19} C. [1 Mark]

• **[Step 2: Numerical Substitution]** Given charge q = -3.2 × 10^{-7} C. Number of electrons n = |q| divided by e = (3.2 × 10^{-7} C) divided by (1.6 × 10^{-19} C) = 2.0 × 10^{12} electrons. Since the polythene piece is negatively charged, electrons are gained by polythene from wool. Mass transferred = n · m e = (2.0 × 10^{12}) × (9.1 × 10^{-31} kg) = 1.82 × 10^{-18} kg (negligibly small). [1 Mark]

• **Key Exam Result:** n = 2.0 × 10^{12} electrons transferred from wool to polythene

7. Coulomb force between charges with distance/medium variation [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula]** In medium with relative permittivity (dielectric constant) K, force F med = F vac divided by K = (one-fourth (0.25)$\pi\varepsilon_0$ K) · (q_1 q_2 divided by r^2). [1 Mark]

• **[Step 2: Calculation]** Given q_1 = 2.0 × 10^{-7} C, q_2 = 3.0 × 10^{-7} C, r = 30 cm = 0.30 m in air (K = 1). F air = (9.0 × 10^9) · [(2.0 × 10^{-7})(3.0 × 10^{-7}) divided by $(0.3)^2$] = (9.0 × 10^9) · (6.0 × 10^{-14} divided by 0.09) = 6.0 × 10^{-3} N (repulsive). If immersed in water (K = 80): F water = F air divided by 80 = (6.0 × 10^{-3} N) divided by 80 = 7.5 × 10^{-5} N. [2 Marks]

• **Key Exam Result:** F air = 6.0 × 10^{-3} N (repulsive); F water = 7.5 × 10^{-5} N